



INSTITUTE OF TECHNOLOGY OF CAMBODIA



Course Enrollment System

Lecture:

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Project Overview



Overview of this projects

A comprehensive platform that enables learners to discover educational courses while supporting teachers in improving their online teaching methods.

- User registration & authentication
- Role-based access control
- Course catalog management
- Student enrollment workflow
- Classroom assignment & scheduling
- Approval processes

Project Requirements

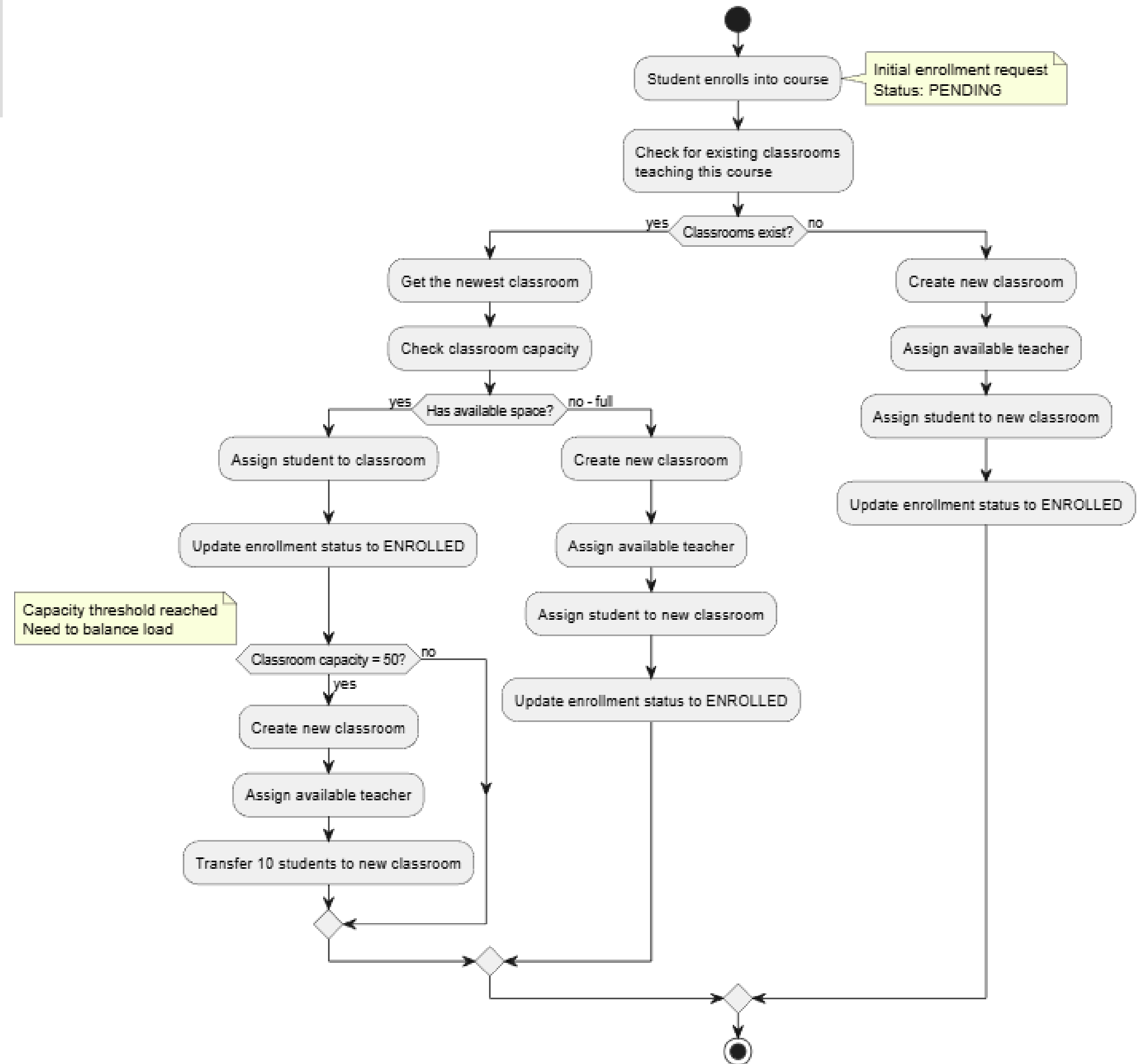
Educational Management System providing:

- Efficient course enrollment process
- Automated classroom scheduling
- Role-based user management
- Scalable web application architecture



Project Flow

Classroom Assignment Flow



Applications & Tools



java



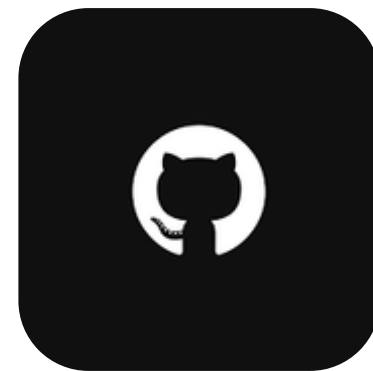
gradle



Springboot



mysql



github



git

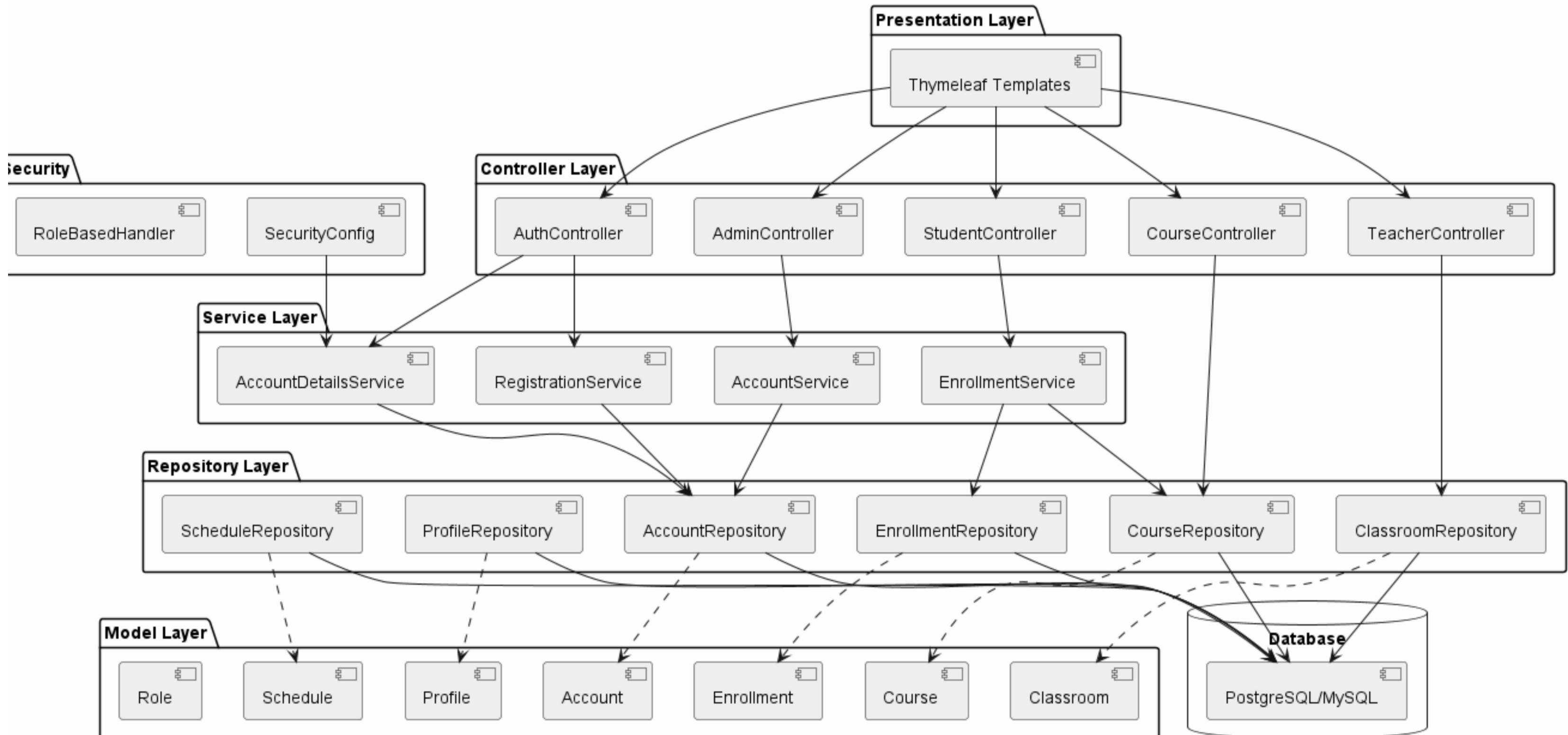
Dependencies

```
3      <!-- Spring Boot Starters -->
4      <dependency>
5          <groupId>org.springframework.boot</groupId>
6          <artifactId>spring-boot-starter-web</artifactId>
7      </dependency>
8
9      <dependency>
10         <groupId>org.springframework.boot</groupId>
11         <artifactId>spring-boot-starter-thymeleaf</artifactId>
12     </dependency>
13
14     <dependency>
15         <groupId>org.springframework.boot</groupId>
16         <artifactId>spring-boot-starter-data-jpa</artifactId>
17     </dependency>
18
19     <dependency>
20         <groupId>org.springframework.boot</groupId>
21         <artifactId>spring-boot-starter-security</artifactId>
22     </dependency>
23
24     <dependency>
25         <groupId>org.springframework.boot</groupId>
26         <artifactId>spring-boot-starter-validation</artifactId>
27     </dependency>
28
29     <dependency>
30         <groupId>org.springframework.boot</groupId>
31         <artifactId>spring-boot-starter-actuator</artifactId>
32     </dependency>
```

```
34     <!-- Database -->
35     <dependency>
36         <groupId>com.mysql</groupId>
37         <artifactId>mysql-connector-j</artifactId>
38         <scope>runtime</scope>
39     </dependency>
40
41     <dependency>
42         <groupId>com.h2database</groupId>
43         <artifactId>h2</artifactId>
44         <scope>runtime</scope>
45     </dependency>
46
47     <!-- Flyway -->
48     <dependency>
49         <groupId>org.flywaydb</groupId>
50         <artifactId>flyway-core</artifactId>
51     </dependency>
52
53     <dependency>
54         <groupId>org.flywaydb</groupId>
55         <artifactId>flyway-mysql</artifactId>
56     </dependency>
57
58     <!-- Thymeleaf Security -->
59     <dependency>
60         <groupId>org.thymeleaf.extras</groupId>
61         <artifactId>thymeleaf-extras-springsecurity6</artifactId>
62     </dependency>
63
```

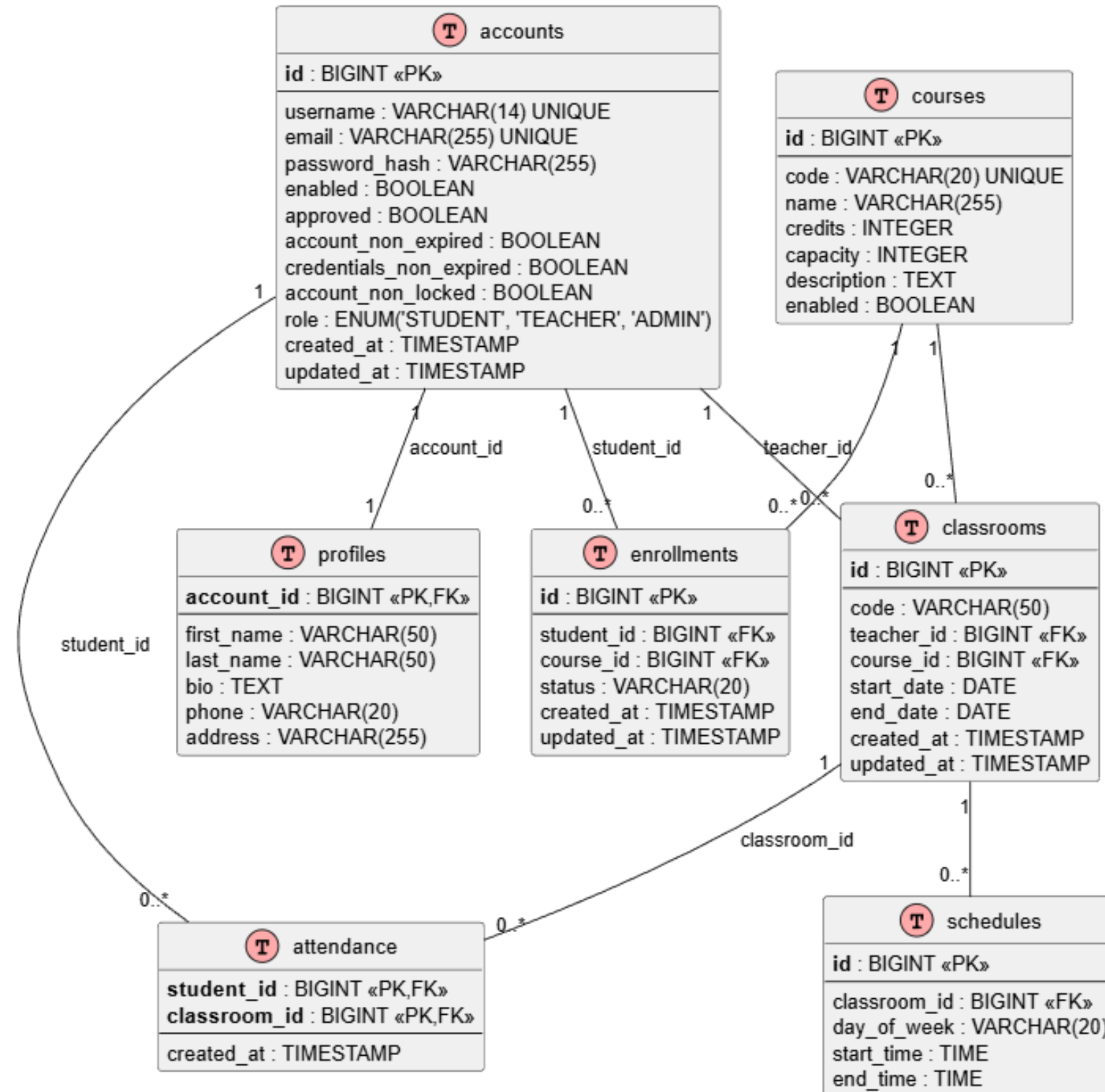

Architecture

Course Enrollment System - Architecture



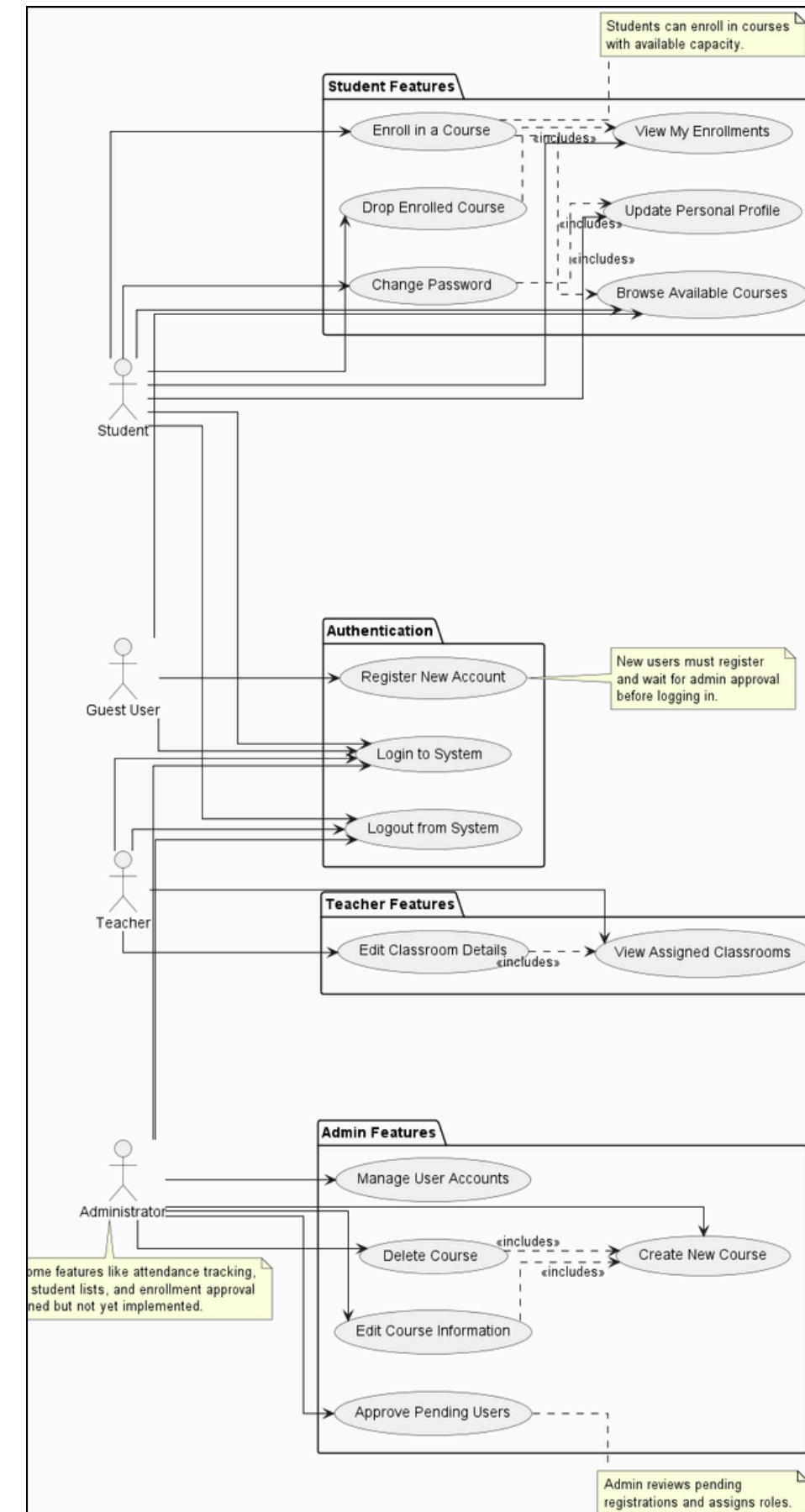
Database Diagram

Course Enrollment System - Database Schema

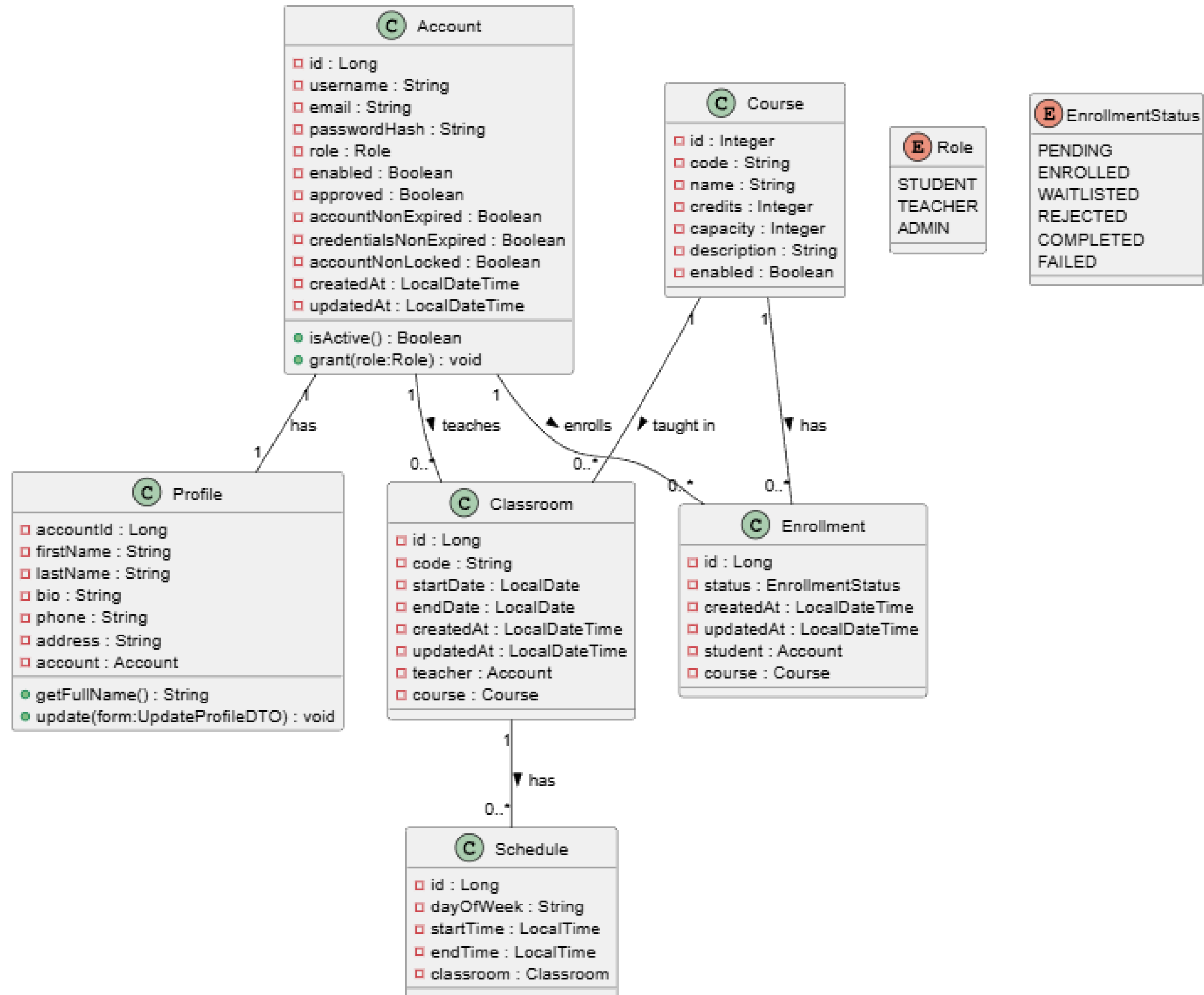


Use case Diagram

```
1 actor "Guest User" as Guest
2 actor "Student" as Student
3 actor "Teacher" as Teacher
4 actor "Administrator" as Admin
5
6 package "Authentication" {
7     usecase "Register New Account" as register
8     usecase "Login to System" as login
9     usecase "Logout from System" as logout
10 }
11
12 package "Student Features" {
13     usecase "Browse Available Courses" as viewCourses
14     usecase "Enroll in a Course" as enroll
15     usecase "View My Enrollments" as myEnroll
16     usecase "Drop Enrolled Course" as drop
17     usecase "Update Personal Profile" as profile
18     usecase "Change Password" as changePass
19 }
20
21 package "Teacher Features" {
22     usecase "View Assigned Classrooms" as myClass
23     usecase "Edit Classroom Details" as editClass
24 }
25
26 package "Admin Features" {
27     usecase "Approve Pending Users" as approveUser
28     usecase "Manage User Accounts" as manageUser
29     usecase "Create New Course" as createCourse
30     usecase "Edit Course Information" as editCourse
31     usecase "Delete Course" as deleteCourse
32 }
```



Entities Diagram



API Endpoints

GET	/login	Displays the login page	AuthController
GET	/register	Displays the user registration form	AuthController
POST	/register	Processes user registration and redirects to login	AuthController

```
@PostMapping("/register")
public String register(
    @Valid @ModelAttribute("registerForm") RegistrationDTO form,
    BindingResult result,
    Model model,
    RedirectAttributes redirection
) {
    if (result.hasErrors()) {
        return "auth/register";
    }

    registrationService.registerUser(form);
    redirection.addFlashAttribute(
        "success",
        "Account registered! Please wait for admin approval before logging in."
    );
    return "redirect:/login";
}
```

API Endpoints

Method	Endpoint	Description	Controller
GET	/admin/dashboard	Displays the admin dashboard with pending user approvals	AdminController
POST	/admin/approvals/{id}/approve	Approves a pending user registration and assigns a role	AdminController
POST	/admin/approvals/{id}/deny	Denies and deletes a pending user registration	AdminController

```
@GetMapping("/dashboard")
public String dashboard(
    Model model,
    @AuthenticationPrincipal AccountDetails accountDetails,
    @RequestParam(defaultValue = "0") int page,
    @RequestParam(defaultValue = "10") int size
) {
    Page<Account> pendingUsers = accountRepository.findByApprovedFalse(
        PageRequest.of(page, size)
    );

    model.addAttribute("pendingUsers", pendingUsers.getContent());
    model.addAttribute("pendingUserCount", pendingUsers.getTotalElements());
    model.addAttribute("currentPage", page);
    model.addAttribute("totalPages", pendingUsers.getTotalPages());
    model.addAttribute("allRoles", Role.values());
    return "admin/dashboard";
}
```

API Endpoints

GET	/classrooms	Lists all classrooms with associated schedules	ClassroomController
GET	/classrooms/{id}/edit	Displays the edit form for a specific classroom (teacher access only)	ClassroomController
POST	{id}/edit	Updates a classroom (note: mapping lacks base path prefix)	ClassroomController

API Endpoints

GET	/courses	Lists all courses with pagination	CourseController
GET	/courses/{id}/enroll	Enrolls the authenticated student in a course	CourseController
POST	/courses/{id}/drop	Drops the authenticated student from a course	CourseController
GET	/courses/create-course	Displays the course creation form	CourseController
POST	/courses/create-course	Creates a new course	CourseController
GET	/courses/edit-course/{id}	Displays the edit form for a specific course	CourseController
POST	/courses/edit-course/{id}	Updates an existing course	CourseController
POST	/courses/delete-course/{id}	Deletes a course	CourseController

```
@GetMapping
public String listCourses(
    Model model,
    @RequestParam(defaultValue = "0") int page,
    @RequestParam(defaultValue = "10") int size
) {
    Page<Course> coursePage = courseRepository.findAll(
        PageRequest.of(page, size)
    );
    model.addAttribute("courses", coursePage.getContent());
    model.addAttribute("currentPage", page);
    model.addAttribute("totalPages", coursePage.getTotalPages());
    return "course/list";
}
```

Conclusion

The Course Enrollment System successfully demonstrates the practical application solving real-world educational challenges, providing a secure, scalable platform that effectively manages the complex workflows of course enrollment, user management, and classroom scheduling.

Project Impact :

- Comprehensive educational management solution
- Scalable enterprise-level application
- Real-world implementation of modern Java technologies
- Foundation for further educational system development

Demo

- Login as student
- View available courses
- Enroll in a course
- View enrolled courses
- Admin manages courses



**Thank
You**